δ-rings and Witt vectors

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Abstract

We give the theory of Witt vectors a new formulation using the methods of universal algebra and category theory. We introduce an algebraic structure, that of a δ-ring. A δ-ring is a commutative ring equipped with an additional unary operation satisfying simple algebraic identities. The main result consists in showing that the forgetful functor from the category of δ-rings to commutative rings has a right adjoint W. An explicit computation allows the identification of $W(A)$ with the ring of Witt vectors over A. We give more conceptual proofs of several results of the theory of Witt vectors [7, 2].

1 δ-rings

Fix once and for all a prime number p and a power $q = p^r$ $(r \geq 0)$.

Definition 1. *A δ-ring A is a commutative ring equipped with a unary operation $\delta : A \to A$ satisfying the following identities:*

1.

$$\delta(x+y) = \delta(x) + \delta(y) - \sum_{i=1}^{q-1} \frac{1}{p}\binom{q}{i} x^i y^{q-i};$$

2.

$$\delta(xy) = x^q\delta(y) + \delta(x)y^q + p\delta(x)\delta(y), \qquad \delta(1) = 0.$$

We shall also say that A is a δ-ring with respect to the pair (p, q). These conditions imply that the unary operation

$$f(x) = x^q + p\delta(x)$$

is a ring endomorphism of A:

$$f(x+y) = f(x) + f(y), \qquad f(xy) = f(x)f(y), \qquad f(1) = 1.$$

We say that f is the Frobenius endomorphism. Conversely, let f be an endomorphism of a commutative ring A without p-torsion. Suppose that for every $x \in A$,

$$f(x) \equiv x^q \pmod{pA}.$$

We obtain a δ-ring structure on A by defining δ as follows:

$$\delta(x) = \frac{1}{p}(f(x) - x^q).$$

Example 2. *The ring $\mathbb{Z}$ admits a unique δ-ring structure since the only endomorphism of $\mathbb{Z}$ is the identity.*

Example 3. *Let L be a Galois extension of a field K. Let $B \subset L$ be a discrete valuation ring and let $A = B/\mathfrak{p}$. Suppose that p is a uniformizer for B and that $A/pA = \mathbb{F}_q$. One can show [1] the existence of a Frobenius substitution $\sigma \in \mathrm{Gal}(L/K)$ characterised by*

1. *$\sigma(B) = B$;*
2. *$\sigma(x) \equiv x^q \pmod{\mathfrak{p}B}$.*

A morphism of δ-rings is a ring homomorphism which preserves δ. We shall also call it a δ-morphism. We note that the Frobenius endomorphism is a δ-morphism.

Theorem 4. *Let $\mathbb{Z}[x_0, x_1, \ldots]$ be the polynomial ring on an infinite sequence of indeterminates. There is a unique δ-ring structure on $\mathbb{Z}[x_0, x_1, \ldots]$ such that $\delta(x_n) = x_{n+1}$ for all n. Equipped with this structure, $\mathbb{Z}[x_0, x_1, \ldots]$ is the free δ-ring on x_0.*

Let $\mathcal{A}$ denote the category of rings and let $\delta\mathcal{A}$ denote the category of δ-rings.

Theorem 5. *The forgetful functor $U : \delta\mathcal{A} \to \mathcal{A}$ has a right adjoint $W : \mathcal{A} \to \delta\mathcal{A}$.*

Category theory [6] brings to light the equivalence between the preceding theorem and the following one.

Theorem 6. *Let $A \to B$ and $A \to C$ be morphisms of δ-rings. Consider the pushout square*

$$\begin{array}{ccc} A & \to & C \\ \downarrow & & \downarrow \\ B & \to & B \otimes_A C \end{array}$$

There is a unique δ-ring structure on $B \otimes_A C$ such that the canonical morphisms $B \to B \otimes_A C$ and $C \to B \otimes_A C$ are δ-morphisms.

One can give a direct proof of these theorems or else use [8, 9]. As a consequence, we have:

Corollary 7. *The functor $W : \mathcal{A} \to \delta\mathcal{A}$ is representable by the ring $\mathbb{Z}[x_0, x_1, \ldots]$. There is a natural bijection*

$$\gamma_A : W(A) \cong A^{\mathbb{N}}.$$

The bijection γ_A is made explicit as follows:

$$W(A) = \mathrm{Hom}_{\delta\mathcal{A}}(\mathbb{Z}[x_0, x_1, \ldots], W(A)) \cong \mathrm{Hom}_{\mathcal{A}}(\mathbb{Z}[x_0, x_1, \ldots], A) \cong A^{\mathbb{N}}.$$

2 Witt vectors

To see that $W(A)$ is isomorphic to the ring of Witt vectors over A, it suffices to use a different description of the free δ-ring on one generator. This description is based on the following result.

Proposition 8. *In the theory of δ-rings, one can define a unique sequence of unary operations $\delta_0, \delta_1, \delta_2, \ldots$ satisfying the identities*

$$f^n(x) = \delta_0(x)^{q^n} + p\,\delta_1(x)^{q^{n-1}} + \cdots + p^n \delta_n(x) \qquad (n \geq 0).$$

The first two terms of the sequence are $\delta_0(x) = x$ and $\delta_1(x) = \delta(x)$.

Proposition 9. *There is a unique δ-ring structure on $\mathbb{Z}[Y_0, Y_1, Y_2, \ldots]$ such that $\delta_n(Y_0) = Y_n$ for all $n \geq 0$. Equipped with this structure, $\mathbb{Z}[Y_0, Y_1, Y_2, \ldots]$ is the free δ-ring on Y_0.*

Theorem 10. *If $q = p$, the ring $W(A)$ is identified with the ring of Witt vectors over A.*

To each element $x \in W(A)$ corresponds a Witt vector $\pi_A(x) \in A^{\mathbb{N}}$. The description of π_A is analogous to that of γ_A but uses the free δ-ring $\mathbb{Z}[Y_0, Y_1, \ldots]$ instead.

The composite endofunctor $\mathcal{A} \to \delta\mathcal{A} \to \mathcal{A}$ is a comonad [6, 3, 2] on $\mathcal{A}$. For simplicity, we still denote this functor by W. There are natural transformations

$$W(A) \xrightarrow{\varepsilon_A} A, \qquad W(A) \xrightarrow{\Delta_A} W(W(A))$$

satisfying associativity and unity conditions. The homomorphism Δ is closely related to the Artin–Hasse exponential [3]. A δ-ring structure on A is equivalent to a coalgebra structure [6] $\alpha : A \to W(A)$. This is described as follows: for every $x \in A$, we have

$$\pi_A(\alpha(x)) = (\delta_0(x), \delta_1(x), \ldots).$$

Remark 11. *Let A be a ring without p-torsion equipped with an endomorphism f such that for every $x \in A$, $f(x) \equiv x^p \pmod{pA}$. The Dieudonné–Cartier lemma [4, page 508] asserts the existence of a homomorphism $s : A \to W(A)$. In the present context, this result is interpreted as follows: the hypotheses imply the existence of a δ-ring structure on A, and one verifies that $s = \alpha$.*

Remark 12. *Under the hypotheses of the preceding remark, we give a particularly simple description of $W(A)$ [5]. Set*

$$B = \{(a_n) \in A^{\mathbb{N}} \mid a_{n+1} \equiv f(a_n) \pmod{p^{n+1}} \ \forall n\}.$$

One verifies that B is a subring of the product ring $A^{\mathbb{N}}$. The translation operator $t((a_n)) = (a_{n+1})$ is an endomorphism of B, and for every $x \in B$,

$$t(x) \equiv x^p \pmod{pB}.$$

Since B is p-torsion-free, we obtain a δ-ring structure on B for which t is the Frobenius endomorphism. One then checks directly that with the projection $B \to A$, the ring B becomes the cofree δ-ring over the ring A, hence $B \cong W(A)$.

References

[1] N. Bourbaki. *Algèbre*, Chap. V, §11.

[2] N. Bourbaki. *Algèbre*, Chap. 8 et 9.

[3] M. Hazewinkel. *Formal Groups and Applications.* Academic Press Monographs 78, 1978.

[4] L. Illusie. Complexe de De Rham-Witt et cohomologie cristalline. *Annales Scientifiques de l'École Normale Supérieure*, 4e série, t. 12, 1979, p. 501–661.

[5] M. Lazard. *Commutative Formal Groups*. Lect. Notes in Math. 443, Springer-Verlag.

[6] S. MacLane. *Category Theory for the Working Mathematician*. Graduate Texts in Math. 5, Springer-Verlag.

[7] P. Ribenboim. *L'Arithmétique des corps*. Hermann, Paris 1972.

[8] D.O. Tall, G.C. Wraith. Representable Functors and Operations on Rings. *Proceedings of the London Mathematical Society* (3) 20 (1970), p. 619–643.

[9] G.C. Wraith. Algebras over theories. *Colloquium Mathematicum*, vol. XXIII (1971), fasc. 2.

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